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Abstract

This project proposes the design, development, construction, and flight test of a high-powered rocket, carrying a deployable glider. A team of five electrical engineers will focus on the development of the rocket and drone while integrating complex electronic embedded systems, controls, communication, data acquisition, and power requirements to carry out the project.

The rocket will carry the deployable payload to a target apogee between 1,500 and 4000 feet above ground level (AGL). Once appropriate flight conditions are met, the control system will autonomously initiate the deployment of the glider, followed by a parachute deployment to safely guide the rocket down.

After apogee deployment, the drone will transition into stable flight. The drone will then be controlled manually from a flight station. Sensors will monitor telemetry throughout the flight; these recordings will asset flight and be recorded or broadcast. A parachute will deploy to facilitate the rockets' recovery. In the event there is a non-viable drone launch, the parachute will also accommodate a drone recovery.

Objective

The Objective is to design, build, integrate, and flight test a high-powered rocket carrying a drone payload demonstrated at the TRA Phoenix launch site.
There are many technical objectives and they include:

- Design the rocket for a target apogee between 1,500 and 4,000 feet using solid rocket motors.
- Use Rocksim to model the expected trajectory, stability, velocity, acceleration, and other essential vehicle characteristics.
- Detect the rockets apogee automatically using sensors.
- Automatically deploy the drone at the rocket’s apogee and then the rockets parachute shortly after.
- Transition from the rocket deployed to a stable drone flight.
- Control the glider/drones' return flight; this will include a controlled downward decent.
- Collect telemetry during the rocket launch and the drone flight to examine performance, deployment, stability, controls and mission success.
- Complete all construction in compliance with the TRA Phoenix launch site requirements.

Introduction

Project RAPTOR (Rocket Apogee Payload Transition to Optimal Return) will be undertaken this year as our team's capstone project. While originally a multidisciplinary project, the lack of interest from the ME department has left the project in the hands of the EE department. The goal of the project is to have a rocket ascend to between approximately 1500-4000 ft. AGL, autonomously detect apogee and ejects a drone that transitions to stable flight and can be controlled back to a designated recovery area. The system will include many aspects. Electrical design including but not limited to components like the flight computer, data acquisition system, power system, and control/navigation system for the glider, refer to figure 3. Mechanical design including but not limited to components like rocket configuration, aerodynamics of both rocket and glider, and the drone wing folding/unfolding system. The first semester will be focused on the design and ordering of parts, with the second focused on refining the project. All things done by the team will need to be done within the safety guidelines of TRA (Tripoli Rocketry Association) and launch site requirements.

Description

This project can be broken up into 3 phases to for readers understanding:

Phase 1: Rocket Launch

Phase 1 will include the rocket launch. An electronic ignition will start a solid rocket motor, propelling the rocket between 1500 and 2500 ft into the air, this is also shown in figure 4. The rocket motor will have to depend on the size and weight of the rocket. The rocket will have a drone inside and will carry it up to apogee. The Rocket design will heavily rely on drone size and desired altitude also refer to figure 1 labeled R.

Phase 2: Rocket and Drone Separation

Phase 2 will include the autonomous deployment of the drone at apogee. This will be done with sensors to detect when the rocket is at zero vertical velocity, the peak of flight. The drone will be deployed and the two will separate, with a parachute deploying for the rocket to safely float back to the ground. Refer to figure 1 labeled 1.

Phase 3: Drone Flight

Phase 3 will be getting control of the drone after deployment and getting it to a designated retrieval zone. Once the two are separated, the drone will need to gain stable control, and from the ground the team will be able to direct it to a certain area of landing. Once an acceptable amount of controlled flight time, a parachute will deploy the drone to safely return to the ground. Refer to figure 1 labeled 2 through 6.

Standards

The project will be under a certain set of standards set by the customer. These standards include the goal, and what is expected throughout the project from the team.

- **Integrated System:** Build a flight-ready rocket with an internally carried/integrated glider.
- **Target Apogee:** Reach 1,500–2,500 ft AGL; model trajectory, stability, and speed in Rocksim.
- **Autonomous Deployment:** Detect apogee autonomously and safely eject the glider.
- **Stable Transition:** Demonstrate a clean, reliable shift from rocket release into glide flight.
- **Active Guidance:** Actively steer the drone back toward a designated recovery zone**.**
- **Telemetry Data:** Record flight performance, dynamics, and data controls for evaluation.
- **Mission Launch:** Execute an integrated flight and recovery at the TRA Phoenix launch site see figure 2.
- **Safety Compliance:** Adhere strictly to Tripoli Rocketry Association (TRA) [1], see figure 2, codes and ground testing.

Figures

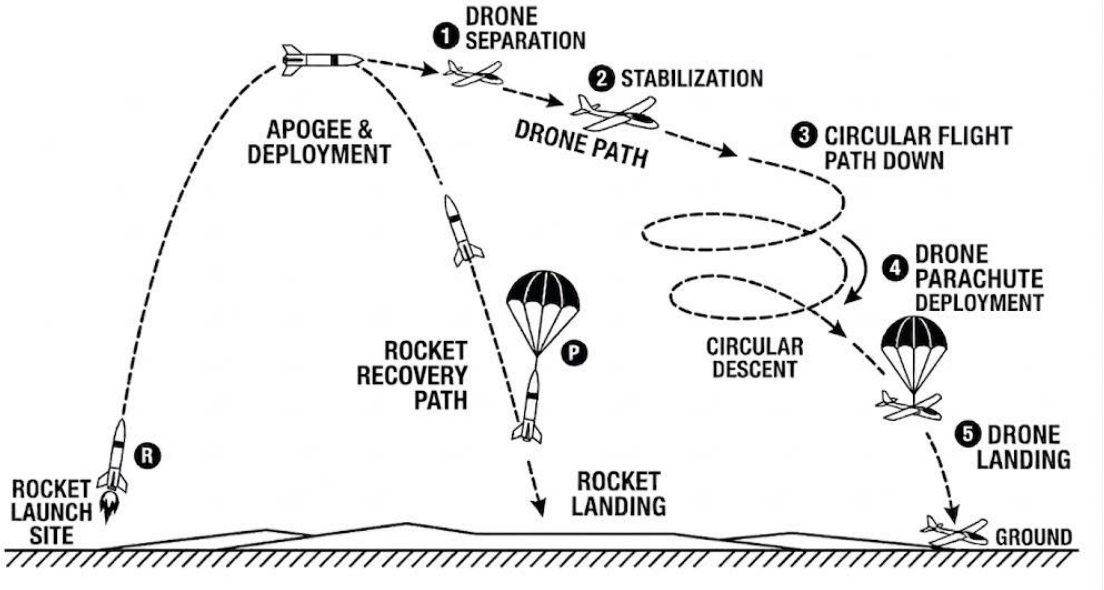


Figure 1 Fight plan for RAPTOR showing the 3 phases.



Figure 2. The Tripoli phoenix logo. [1]

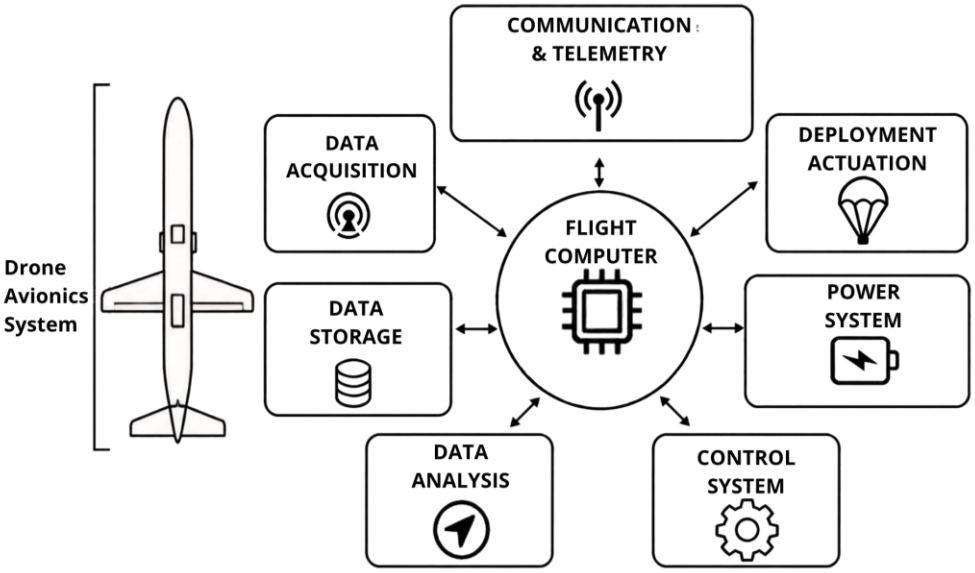


Figure 3. diagram of the drone Avionics system.

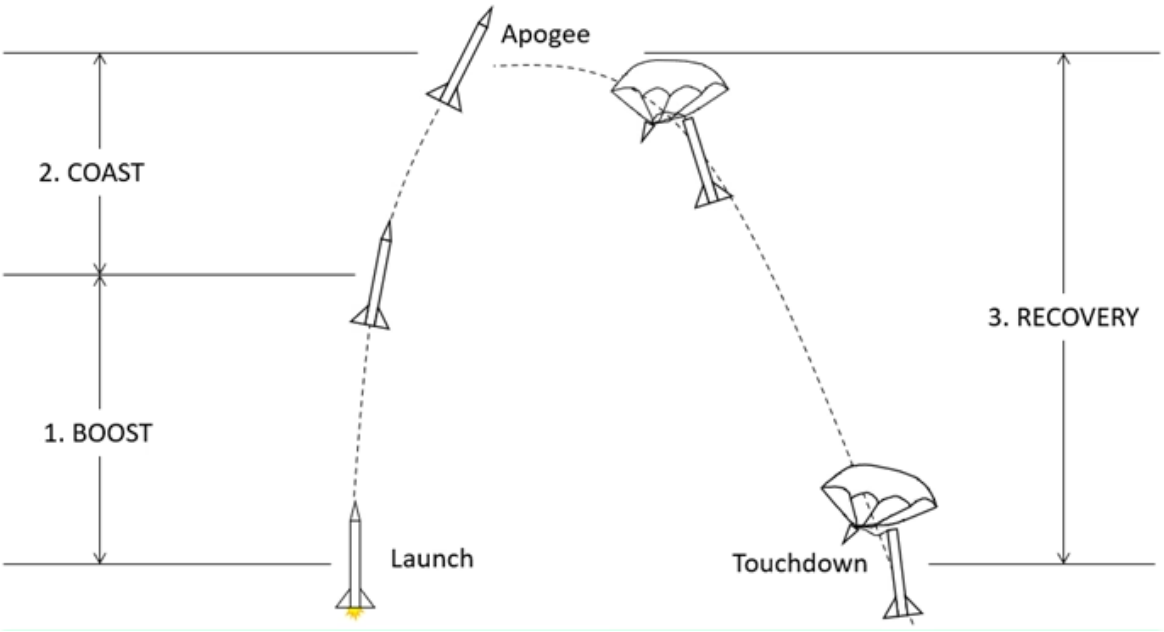


Figure 4. Rocket Flight Phases. [2]

Schedule

The project will be nearly a year long and will be guided by a schedule. Since this project is in its infancy the schedule is quite general, as we do not know all the specifics as of now.

September 2026

- Project creation and approval
- Initial Research

October 2026

- Research
- Mid-term presentation
- Part ordering
- Initial designs

November 2026

- Part receiving
- Research
- Designing / refining designs
- Presentation of videos
- NAU Fest

December 2026

- Final designs
- Initial builds
- Final part receiving
- Final Presentation

January 2027

- Initial testing and evolution
- Initial new designs
- New builds

February 2027

- Final designs
- New builds
- Testing and evaluation

March 2027

- Design refining
- Components / systems refining
- Mid-term presentations

April 2027

- Final build and testing

May 2027

- Final presentation

Conclusion

In conclusion, this year long project entails a lot of work. There will be electronic, aerodynamic and mechanical designs involved and will all need to work together well to prevent catastrophic failure. The work will be done on a schedule throughout the year and will need all hands-on deck to complete the various difficult tasks that are sure to be involved. The team will be in constant communication with the customer to ensure a timely and correct product by the end of the capstone.

References

[1] Tripoli Phoenix, "Tripoli Phoenix Home," Accessed 16 Sept. 2026 Available: https://traphx.clubexpress.com/

[2] B. Finio, "Solid Motor Rocket Propulsion," Science Buddies, 9 January 2026. Accessed 16 Sept. 2026 Available: https://www.sciencebuddies.org/science-fair-projects/project-ideas/Phys_p008/physics/model-rocket-propulsion